

Deep Learning–Based Joint Channel Estimation and CSI Feedback for RIS–Assisted Communications

JDCNet: A Unified Deep Learning Approach

Course/Project: ML for Wireless Communication

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Reconfigurable Intelligent Surfaces (RIS)

RIS Technology

Programmable reflecting surface, many elements N , can program phases to shape wireless channels.

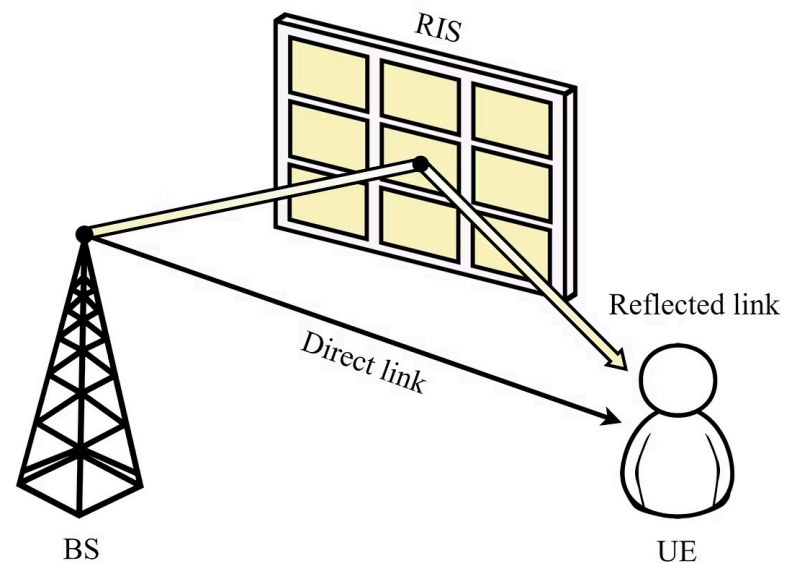


CSI Requirement

Accurate channel state information needed for optimal beamforming.

Feedback Necessity

CSI must be estimated at UE and fed back to BS.



The Core Research Problem: Optimising CSI for RIS

How can we efficiently acquire and feedback CSI in RIS-assisted systems while minimizing pilot overhead, feedback bandwidth, computational load, and estimation errors?

Key Challenges:

Dimensionality and Pilot

Very high-dimensional CSI (BS antennas M $\times$ RIS elements N). Pilot and feedback costs explode with the number of RIS elements (N), leading to inefficiency.

Feedback & Quantisation:

Managing the immense feedback overhead and quantisation errors inherent in high-dimensional CSI.

Error Propagation:

Addressing the compounding effect of errors, from initial estimation through compression and final reconstruction.

Literature Survey: Existing Approaches

Channel Estimation Methods

- **SRNet:** Super-resolution network for fine-grained channel estimation
- **CNN-based:** Convolutional neural networks for spatial feature extraction
- **EDSR:** Enhanced deep super-resolution for channel reconstruction

CSI Feedback Methods

- **CsiNet:** Deep learning framework for compressed CSI feedback
- **CLNet:** Convolutional-LSTM network for temporal correlation
- **DCRNet:** Deep complex residual network for improved reconstruction

📌 **Key Gap:** Existing methods address channel estimation and CSI feedback as separate, independent components.

Limitations of Previous Work

Separate Optimization

Channel estimation and CSI feedback modules are designed independently, missing opportunities for joint optimization.

Error Propagation

Estimation errors propagate directly to the feedback stage, compounding performance degradation.

Sparsity Assumption

Traditional compressive sensing methods rely on channel sparsity assumptions that may not hold in practical RIS scenarios.

Feedback Overhead

Deep learning methods still suffer from high feedback overhead despite improved reconstruction accuracy.

Conclusion: These limitations motivate the need for **joint optimization of channel estimation and CSI feedback** through unified deep learning architecture.

Proposed Solution: JDCNet

JDCNet — Joint Deep Learning Channel Estimation and CSI Feedback Network

1

Joint Optimization

Simultaneously optimizes both channel estimation and feedback compression

2

Encoder–Decoder Architecture

Deep learning framework with complementary encoder at UE and decoder at BS

3

Reduced Overhead

Significantly reduces feedback bits while maintaining reconstruction accuracy. Keep UE encoder lightweight

JDC-Net Architecture: Encoder–Decoder Framework

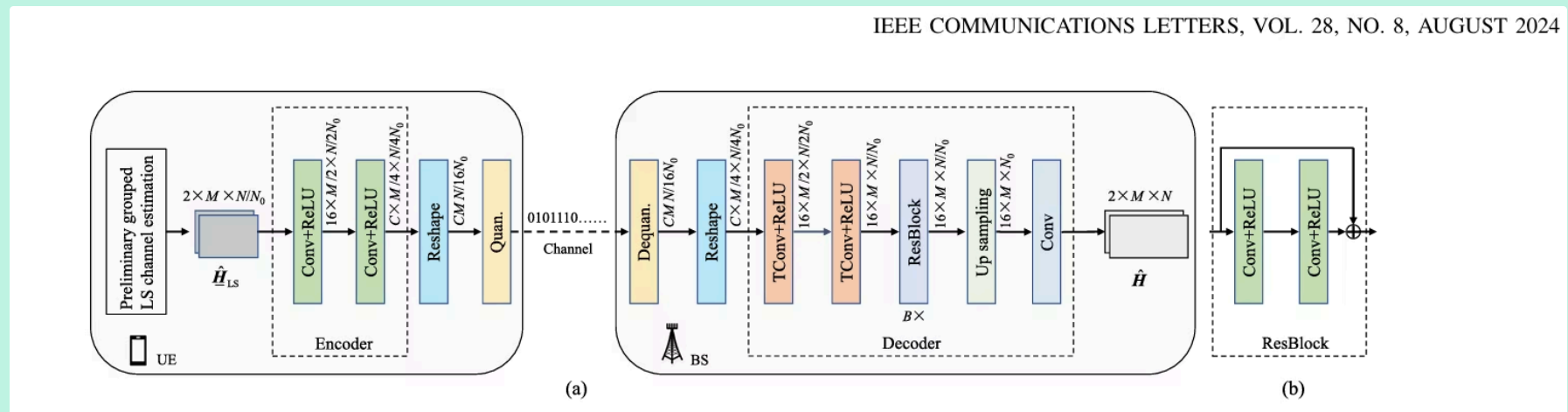
Encoder (UE Side)

- Processes preliminary least-squares (LS) channel estimate
- CNN layers extract spatial and spectral features
- Feature vector compressed through dimensionality reduction
- Quantization for efficient bit representation

Decoder (BS Side)

- Receives compressed feedback bits from UE
- Transposed convolutional layers for up-sampling
- Residual blocks for refined reconstruction
- Output: Full channel matrix reconstruction

□ Network architecture figure showing encoder–decoder layers



System Model: Communication Setup

1

Base Station (BS)

Equipped with **M antennas** for multi-antenna transmission and reception

2

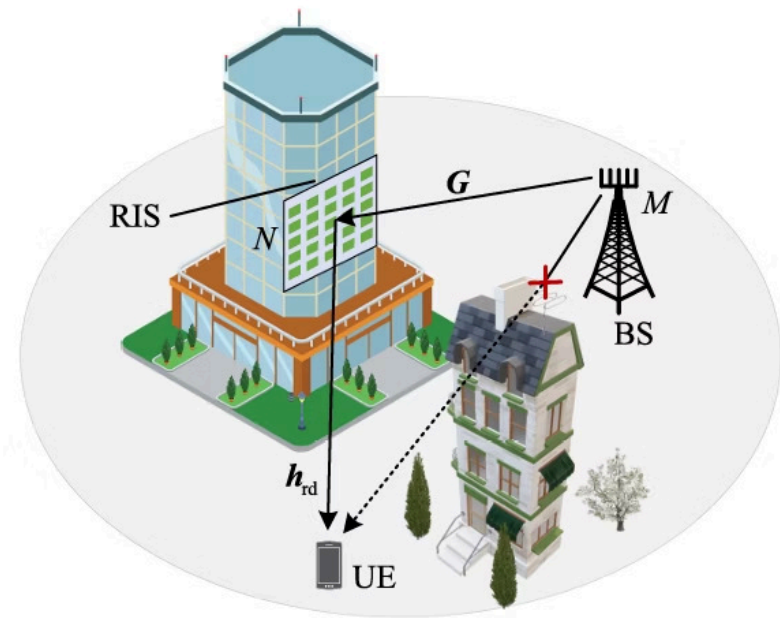
Reconfigurable Intelligent Surface (RIS)

Contains **N passive reflecting elements** for programmable signal control

3

User Equipment (UE)

Single-antenna device for receiving RIS-reflected signals.



$$\mathbf{G} \in \mathbb{C}^{N \times M} : BS \rightarrow RIS \text{ channel}$$
$$\mathbf{h}_{ru} \in \mathbb{C}^{N \times 1} : RIS \rightarrow UE \text{ channel}$$

Channel Model

Communication occurs exclusively through **RIS reflection path**. No direct path.

Cascaded Channel

Effective channel comprises **BS $\rightarrow$ RIS** and **RIS $\rightarrow$ UE** segments in series.

Mathematical Model of the RIS-Assisted System

M-antenna Base Station, an N-element Reconfigurable Intelligent Surface (RIS), and a single-antenna user.

Channel Matrices

Matrix	Description	Dimensions
$\mathbf{G}$	BS to RIS	$N \times M$
$\mathbf{h}_{rd}$	RIS to UE	$N \times 1$
$\mathbf{\Omega}$	Reflection	$N \times N$

RIS Reflection Matrix

$$\mathbf{\Omega} = \text{diag}(\phi_1, \phi_2, \dots, \phi_N)$$

Received Signal at UE

$$y(t) = \mathbf{h}_{rd}^T \mathbf{\Omega} \mathbf{G} \mathbf{w}(t) s(t) + z(t)$$

Cascaded Channel Formulation

$$\mathbf{H} = \text{diag}(\mathbf{h}_{rd}^T) \mathbf{G}$$

Simplified Received Signal

$$y(t) = \boldsymbol{\phi}^T(t) \mathbf{H} \mathbf{w}(t) s(t) + z(t)$$

Number of Channel Parameters

$$M \times N$$

Example Scenario

If $M = 16$ and $N = 64$, total parameters = 1024.

Dimension Reduction: The Grouping Approach

Grouping Strategy

- Divide RIS into groups of size N_0 (adjacent elements combined). Number of groups: $N' = N/N_0$.
- Grouped effective channel reduces unknowns from $M \times N$ to $M \times N'$.

Key Benefits

Reduced Pilot Overhead

Requires fewer symbols for accurate estimation.

Lower Feedback Bandwidth

Decreases data needed for link adaptation.

Faster Convergence

Accelerates deep learning training speeds.

Vectorized Channel Model

$$y(t) = (\boldsymbol{\omega}^T(t) \otimes \boldsymbol{\phi}^T(t)) \text{vec}(\mathbf{H}) + z(t)$$

- $\text{vec}(\mathbf{H})$: Vectorized cascaded channel
- $\otimes$: Kronecker product

Simplified Form

$$\begin{aligned} y &= (\mathbf{V} \otimes \boldsymbol{\Phi}) \text{vec}(\mathbf{H}) + \mathbf{Z} \\ &= \mathbf{W} \text{vec}(\mathbf{H}) + \mathbf{Z} \end{aligned}$$

Channel Estimation and CSI Feedback

LS Channel Estimation

$$\text{vec}(\hat{\mathbf{H}}_{\text{LS}}) = (\mathbf{W}^H \mathbf{W})^{-1} \mathbf{W}^H \mathbf{y}$$

Where $\mathbf{W} = \mathbf{V} \otimes \Phi$ is the matrix of beamforming vectors and RIS phase shifts.

CSI Compression

Compression ratio γ :

$$\gamma = \frac{CMN/16N_0}{2MN} = \frac{C}{32N_0}$$

Encoder Operation at UE

The encoder at the UE compresses the LS channel estimate and obtains a feature vector $\mathbf{v}$ of dimension $CMN/16N_0$:

$$\mathbf{v} = \text{Flatten}(f_{\text{encode}}(\hat{\mathbf{H}}_{\text{LS}}))$$

where $f_{\text{encode}}()$ and $\text{Flatten}()$ represent encoder compression and flattening operations, respectively.

Initial Results: Experimental Evaluation

Performance Metric

Normalized Mean Square Error (NMSE) used to quantify channel reconstruction accuracy:

$$\text{NMSE} = \frac{\|\mathbf{H} - \hat{\mathbf{H}}\|_F^2}{\|\mathbf{H}\|_F^2}$$

where $\mathbf{H}$ is the true channel and $\hat{\mathbf{H}}$ is the reconstructed channel.

Key Observations

- NMSE decreases as **SNR increases** for all methods
- Deep learning methods **outperform compressive sensing** approaches
- JDCNet achieves **lowest NMSE across all SNR values**

Performance Analysis: Grouping & RIS Elements

Dataset Generation

- ❏ The dataset was generated using MATLAB-based wireless channel simulation tool called SimRIS and exported as .mat files, which were later used in the Py-Torch training pipeline.

The screenshot displays the SimRIS simulation tool interface, which is divided into several sections for configuring simulation parameters and locations.

Simulation Parameters:

- Environment: Indoor (InH Office)
- RIS Position: xz plane (Scenario 1)
- Frequency: 28 GHz
- Array Type: Uniform Linear Array
- Number of Transmit Antennas (Nt):
- Number of Receive Antennas (Nr):
- Number of RIS elements (N):
- Number of Channel Realizations:

Locations:

- Recommended
- Transmitter Coordinates: X: 0, Y:, Z:
- Rx Coordinates: X:, Y:, Z:
- RIS Coordinates: X:, Y:, Z:

Error Control:

Run SimRIS

Save as

CoreLab Website

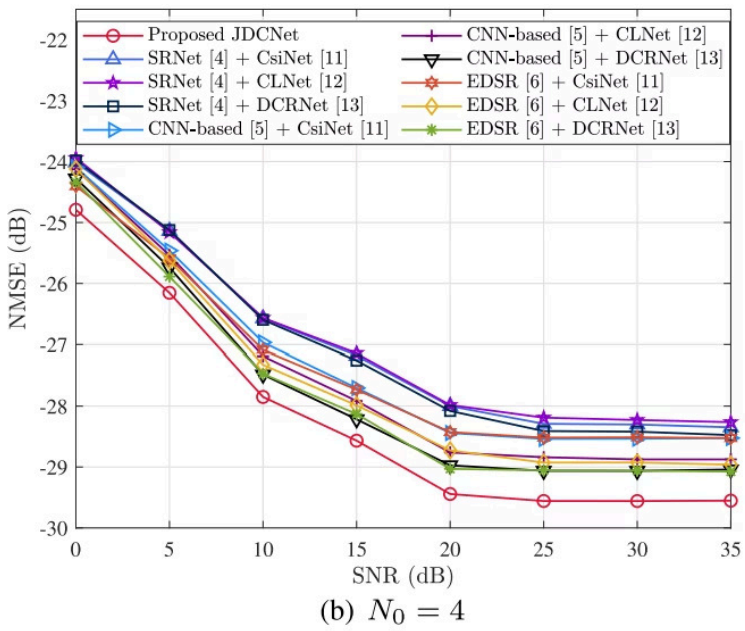
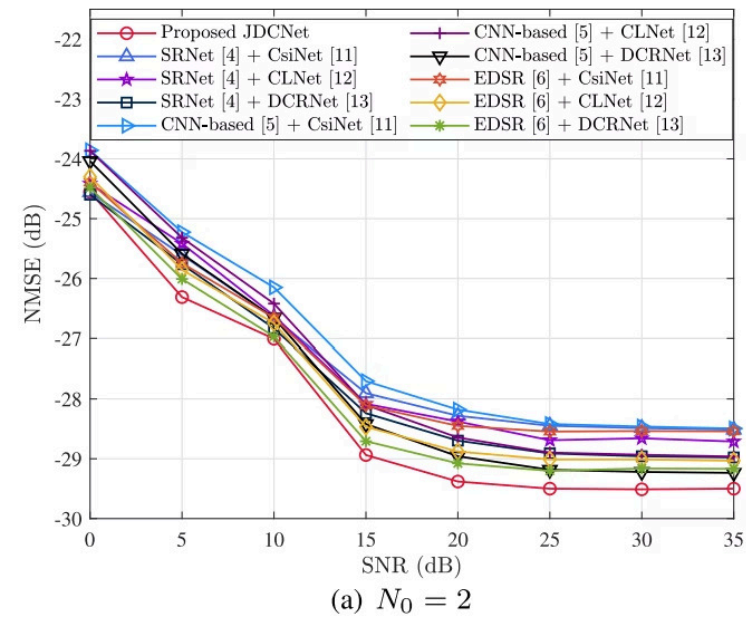
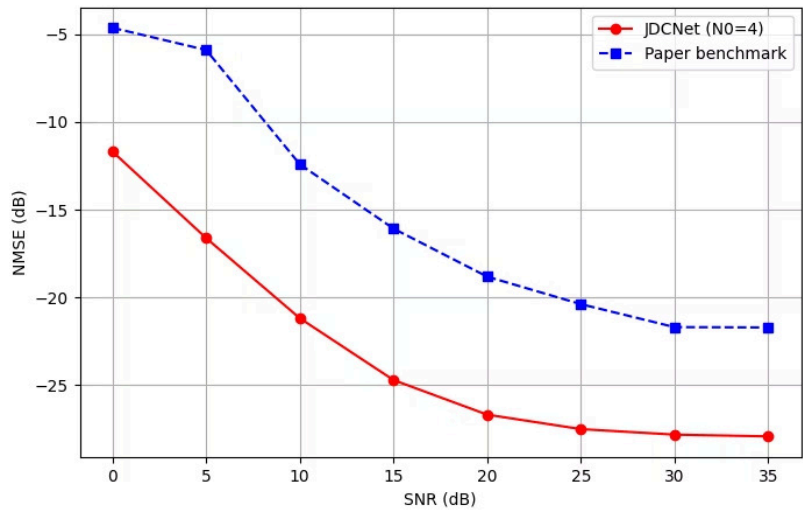
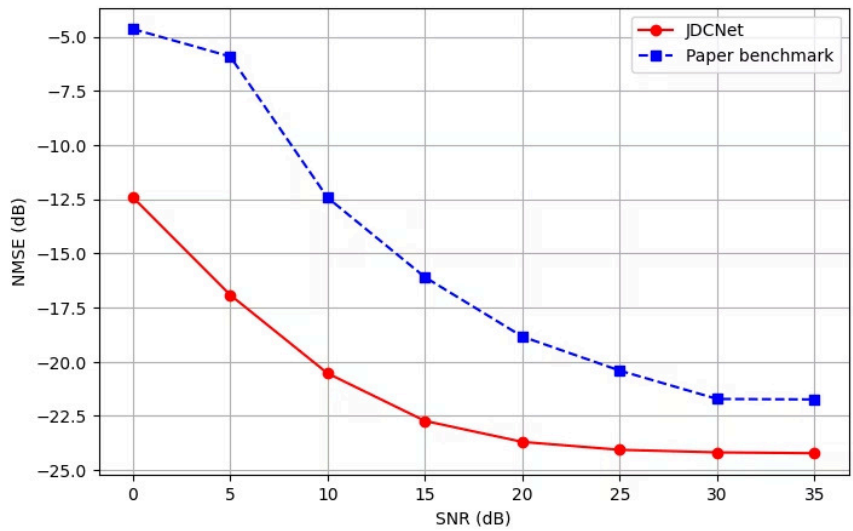
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Simulation Results

PERFORMANCE OF NON-DEEP LEARNING BENCHMARK METHODS IN TERMS OF NMSE (dB)

SNR (dB)	0	5	10	15	20	25	30	35
TAVL3 [21] + BM3D-PRGAMP [22]	-4.64	-5.91	-12.43	-16.10	-18.84	-20.41	-21.72	-21.74



Conclusion

Key Achievements

- **JDCNet:** Joint optimization of channel estimation and CSI feedback
- Reduces error propagation compared to separate methods
- Delivers lower NMSE and improved reconstruction accuracy
- Lowers feedback overhead and computational complexity
- Demonstrates practical feasibility for RIS systems

Timeline

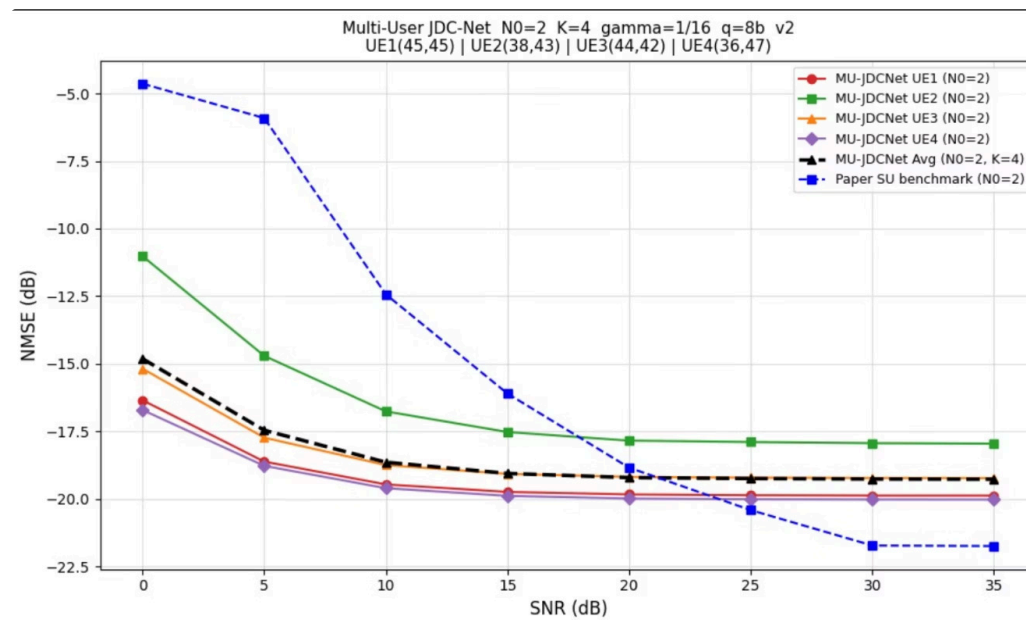
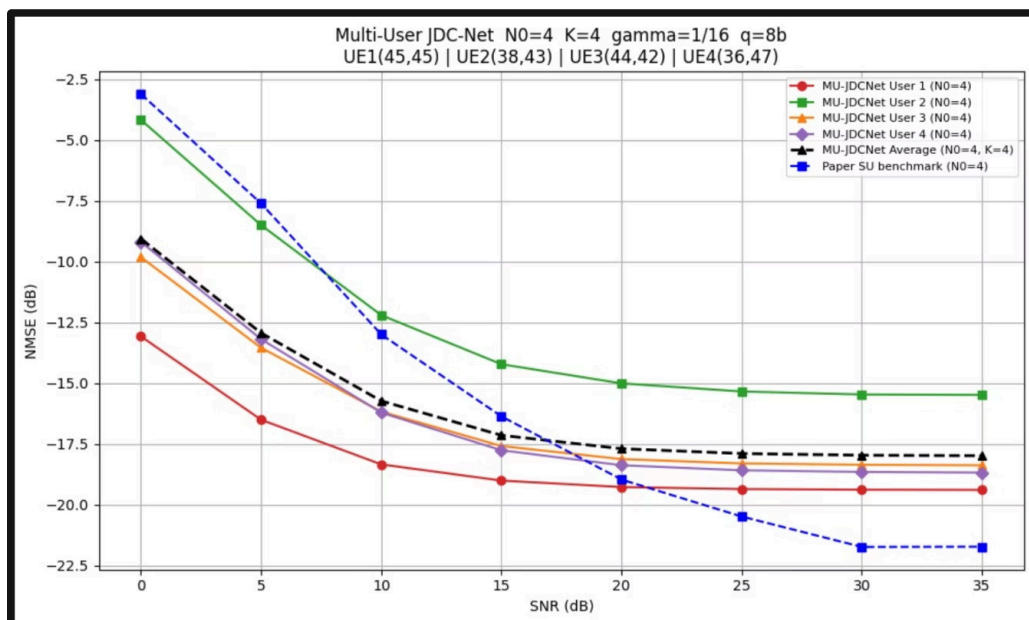
Completed

- Proposed the JDCNet architecture
- Demonstrated the benefits of joint optimization
- Validated NMSE improvements over baselines

Future Work

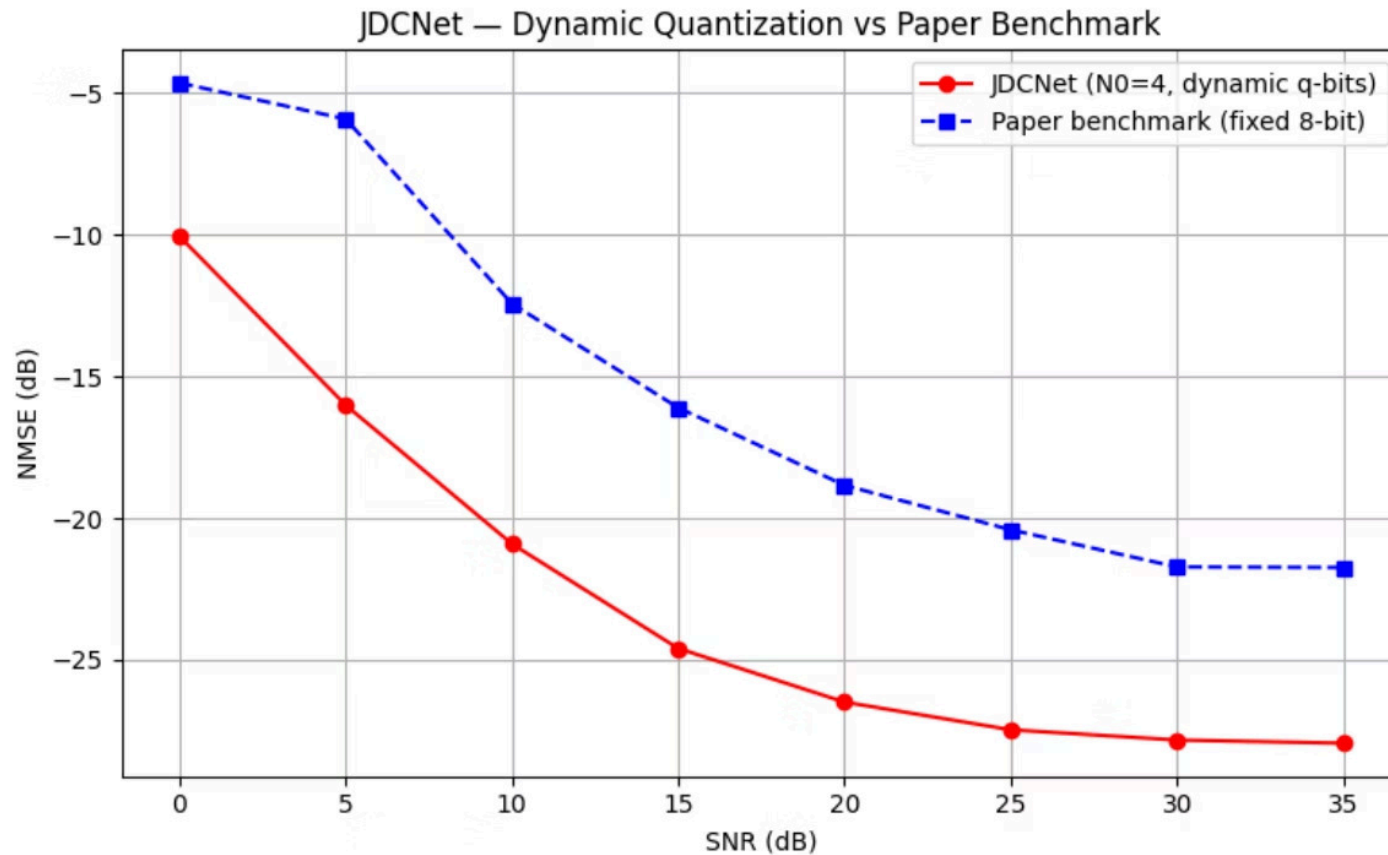
- Extend JDC-Net to jointly estimate and feed back CSI for multiple users
- Adapt quantization bits per sample or channel based on channel quality and SNR
- Reduce computational cost

Multi-User JDC-Net Extension



Initial Dynamic Quantization Attempt (v1)

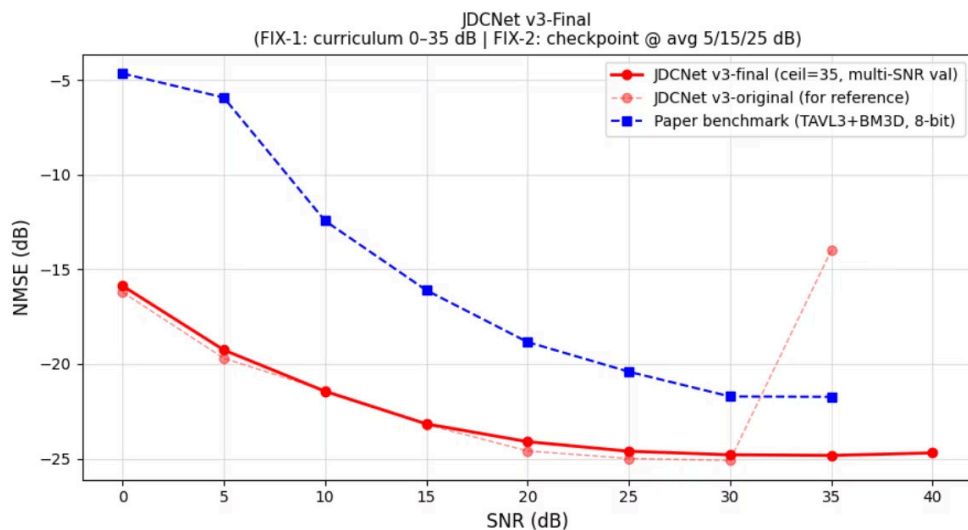
Adapts quantization bits to channel conditions for more efficient feedback.



Dynamically allocates quantization bits based on channel quality and SNR.

Initial results show no improvements over a baseline JDC-Net which we did earlier.

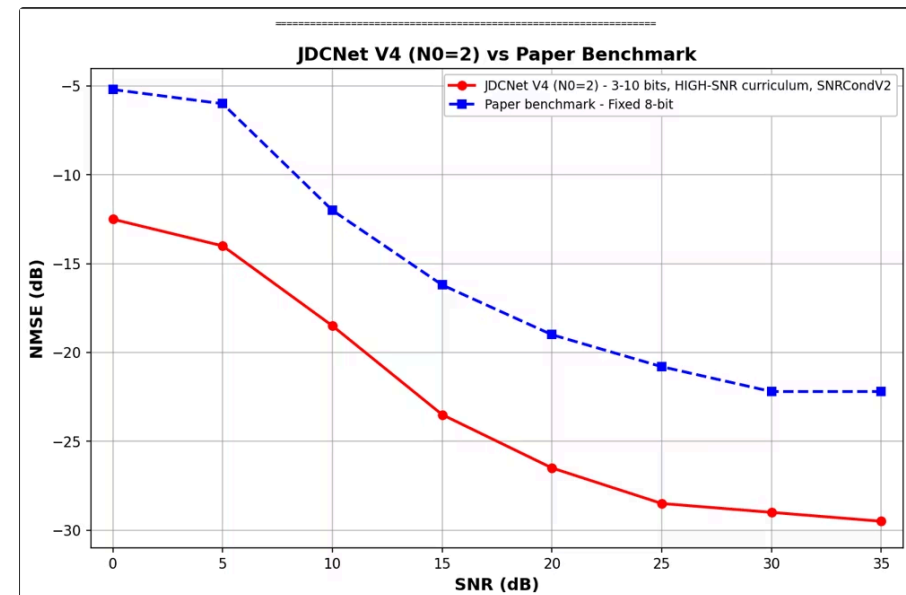
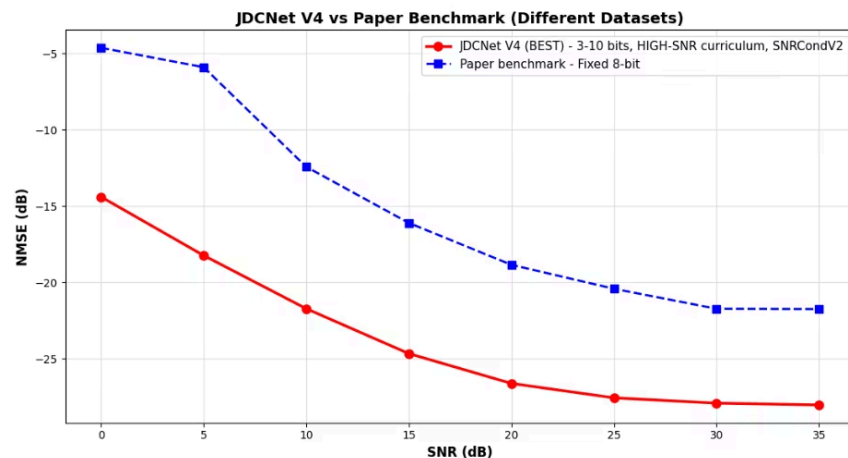
v2 Architecture & Training Improvements



- SNR-conditioned decoding via FiLM layers
- 6-block residual architecture for better feature extraction
- Curriculum training for progressive noise level introduction
- Dynamic bit allocation for rate-distortion optimization
- Gradient clipping and cosine LR with warmup for stable convergence

Improvements observed at lower SNR; comparable performance at higher SNR

Final Architecture Evolution



Final Version Improvements

- Larger MLP-based SNR Conditioner for richer noise-level representations
- Extended quantization range (3–10 bits)
- High-SNR focused curriculum
- Balanced performance across full SNR spectrum
- Consistent improvements at both low and high SNR

Thank You!

Questions are appreciated, if any.